

Torque and Static Equilibrium

Understanding rotational forces and the conditions that keep objects perfectly still

Why can a child on a seesaw balance an adult simply by sitting farther from the center? Why does a longer wrench make a bolt easier to turn? Why does a ladder lean against a wall without sliding — until suddenly it does? These questions all have the same answer: torque. This chapter introduces the rotational equivalent of force and explores the conditions that keep objects in perfect balance.

20.1 Rotation and the Need for a New Concept

In Part II you learned that a net force causes an object to accelerate — to change its linear velocity. But forces can do something else too: they can cause an object to rotate. Push on the edge of a door and it swings. Push on its center and it moves differently. Apply force at different distances from a hinge and the result changes dramatically, even if the force itself is identical.

Linear mechanics cannot fully describe this behavior. The location where a force is applied matters, and so does its direction relative to the object's pivot. To handle rotation, physics introduces a new concept: torque. Torque is to rotation what force is to linear motion. Just as a net force causes linear acceleration, a net torque causes rotational acceleration — or, if balanced, rotational equilibrium.

The transition from linear to rotational thinking is one of the most important conceptual shifts in introductory physics. It requires you to think not just about how strong a force is, but about where it acts and in what direction. A small force applied far from a pivot can have far more rotational effect than a large force applied close in.

20.2 What Torque Is

Torque (symbol: Greek letter tau, τ) is the rotational effect of a force applied at a distance from a pivot point. The pivot — also called the axis of rotation or fulcrum — is the point about which rotation occurs. A door hinge, the axle of a wheel, the elbow joint of an arm: these are all examples of pivot points.

Torque depends on three things: the magnitude of the force, the distance from the pivot to where the force is applied (called the moment arm or lever arm), and the angle at which the force is applied relative to the line connecting the pivot to the point of application.

Conceptually, torque is greatest when the force is applied far from the pivot and directed perpendicular to the lever arm. It is zero when the force points directly toward or away from the pivot, no matter how large the force is, because such a force produces no tendency to rotate.

Torque = Force × Lever Arm × sin(angle between force and lever arm)

$$\tau = F \cdot r \cdot \sin \theta$$

When force is perpendicular to the lever arm, $\sin 90^\circ = 1$, giving maximum torque.

When force points along the lever arm, $\sin 0^\circ = 0$, giving zero torque.

The SI unit of torque is the newton-meter (N·m). Notice that this has the same units as energy (joules), but torque and energy are fundamentally different quantities — torque is a tendency to cause rotation, not a stored capacity for doing work.

Torques have direction — they can cause clockwise or counterclockwise rotation. By convention, counterclockwise torques are treated as positive and clockwise torques as negative (though you may choose either convention as long as you are consistent throughout a problem).

20.3 The Lever Arm

The lever arm (also called the moment arm) is the perpendicular distance from the axis of rotation to the line of action of the force. This definition is crucial and often misunderstood.

The line of action is an imaginary line extending infinitely along the direction of the force. The lever arm is the shortest distance from the pivot to this line — always a perpendicular distance. When you use a wrench to tighten a bolt, the lever arm is not simply the length of the wrench; it is the component of that length measured perpendicular to the direction you are pushing.

This is why the formula $\tau = F \cdot r \cdot \sin \theta$ includes the sine term: it automatically extracts the perpendicular component. An equivalent way to write torque is $\tau = F \cdot d$, where d is the lever arm defined as the perpendicular distance described above. Both expressions give the same result.

Practical implication: to maximize torque for a given force, position your force so that it acts perpendicular to the lever. A force applied at an angle wastes some of its rotational effectiveness on a component that merely pushes or pulls along the lever without causing rotation.

20.4 Calculating Torque

To calculate the torque produced by a force about a given pivot, follow this procedure:

1. Identify the pivot point (axis of rotation).
2. Identify the point where the force is applied and measure the distance r from the pivot to that point.
3. Determine the angle θ between the force vector and the line connecting the pivot to the application point.
4. Calculate $\tau = F \cdot r \cdot \sin \theta$.
5. Assign a sign: positive for counterclockwise tendency, negative for clockwise (or choose your own convention and apply it consistently).

Worked example: A 40 N force is applied to a wrench at a distance of 0.25 m from the bolt. The force is directed at 60° to the wrench handle. Torque = $40 \times 0.25 \times \sin 60^\circ = 40 \times 0.25 \times 0.866 = 8.66 \text{ N}\cdot\text{m}$. If the same force were applied perpendicular to the handle, the torque would be $40 \times 0.25 \times 1 = 10 \text{ N}\cdot\text{m}$ — about 15% more effective.

This example illustrates why good technique matters in physical tasks: a perpendicular push on a wrench maximizes the torque you deliver for a given effort.

20.5 The Principle of Moments

When multiple forces act on a rigid object, each one may produce a torque about a given pivot. The principle of moments (also called Varignon's theorem) states that the total, or net, torque about any pivot is simply the algebraic sum of the individual torques. Torques in the same rotational direction add; torques in opposite rotational directions subtract.

This principle is why a seesaw balances when the torques on each side are equal and opposite. A heavier child sitting close to the center can be balanced by a lighter child sitting farther away, as long as the product (weight $\times$ distance) is the same on both sides.

For a balanced seesaw:

$$F_1 \cdot d_1 = F_2 \cdot d_2$$

(weight $\times$ distance from pivot on left) = (weight $\times$ distance from pivot on right)

The principle of moments allows us to solve many practical problems: finding unknown forces, locating where a force must be applied to achieve balance, or determining whether a structure will remain stationary or begin to rotate.

20.6 Conditions for Static Equilibrium

An object is in static equilibrium when it is completely at rest and has no tendency to start moving or rotating. For static equilibrium to exist, two independent conditions must be satisfied simultaneously:

First condition — Translational equilibrium: The net force on the object must be zero. This means all forces in every direction must cancel out. Equivalently, the sum of all forces in the horizontal direction equals zero, and the sum of all forces in the vertical direction equals zero.

$$\left| \begin{array}{l} \Sigma F_x = 0 \\ \Sigma F_y = 0 \end{array} \right.$$

Second condition — Rotational equilibrium: The net torque about any chosen pivot must be zero. This means all clockwise torques must exactly balance all counterclockwise torques, regardless of which point you choose as the pivot.

$$\left| \Sigma \tau = 0 \quad (\text{about any pivot point}) \right.$$

Both conditions are necessary. An object can satisfy the first condition without the second — think of a pair of equal and opposite forces applied at different points on a body, which produces no net force but does produce a net torque, causing rotation. Similarly, an object can have zero net torque but still accelerate linearly if forces are unbalanced.

The freedom to choose any point as the pivot in the second condition is enormously useful. A well-chosen pivot (typically where an unknown force acts) can eliminate that unknown from the torque equation, simplifying the algebra significantly.

20.7 Choosing the Pivot Wisely

A common source of difficulty in equilibrium problems is not knowing where to take torques. The answer is: you can take torques about any point, and if the object is in equilibrium the result will always be zero. But some choices make the algebra much easier than others.

The golden rule: choose the pivot at the location where an unknown force acts. That force then has a lever arm of zero, so it contributes zero torque and disappears from the equation entirely. This eliminates one unknown and often makes the problem immediately solvable.

For example, consider a horizontal beam supported at its left end by a pin and at its right end by a cable. If you want to find the cable tension, take torques about the pin. The pin reaction forces have zero lever arm (they act at the pivot itself) and vanish from the equation. You are left with only the known weights and the unknown cable tension — one equation, one unknown.

If you later want the pin reaction forces, use the translational equilibrium equations (sum of forces = 0 in each direction), which give you the remaining unknowns directly.

20.8 Solving Static Equilibrium Problems

Static equilibrium problems follow a reliable procedure. Mastering this procedure will allow you to solve a wide variety of problems involving beams, ladders, bridges, cranes, and biological levers.

6. Draw a clear free-body diagram showing all forces acting on the object, including their points of application.
7. Set up a coordinate system. Identify all horizontal and vertical force components.
8. Choose a pivot point (ideally where an unknown force acts).
9. Write the torque equation: $\Sigma\tau = 0$. Include all torques with their correct signs.
10. Write the force equations: $\Sigma F_x = 0$ and $\Sigma F_y = 0$.
11. Solve the system of equations for the unknowns.
12. Check: do the signs make physical sense? Are the magnitudes reasonable?

Common mistake: forgetting that the weight of the beam itself acts at its center of mass (typically its midpoint if the beam is uniform). This weight must be included in both the force and torque equations.

20.9 Levers and Mechanical Advantage

A lever is a rigid rod or beam that rotates about a fixed pivot (fulcrum) and is used to exert a force. Levers are among the oldest and most universal simple machines. They appear in crowbars, scissors, tweezers, wheelbarrows, bottle openers, and the bones of the human body.

The mechanical advantage of a lever is the ratio of the output force (the load you lift or move) to the input force (the effort you apply). From the torque balance condition, the mechanical advantage equals the ratio of the lever arms: $MA = d_{\text{effort}} / d_{\text{load}}$.

There are three classes of lever, distinguished by the relative positions of the pivot, the effort, and the load. In a first-class lever (e.g., a seesaw or crowbar), the pivot is between the effort and the load. In a second-class lever (e.g., a wheelbarrow), the load is between the pivot and the effort. In a third-class lever (e.g., tweezers or the forearm), the effort is between the pivot and the load.

Lever Class	Pivot Position	MA > 1?	Example
First class	Between effort and load	Can be > or < 1	Seesaw, crowbar
Second class	Load between pivot and effort	Always > 1	Wheelbarrow, nutcracker

Third class	Effort between pivot and load	Always < 1 (speed advantage)	Tweezers, forearm
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Table 20.1 The three classes of lever and their mechanical properties.

20.10 Torque in the Human Body

The human musculoskeletal system is a sophisticated network of levers. Bones are the rigid beams, joints are the pivots, and muscles supply the forces. Understanding torque explains a great deal about how the body moves and why it is designed the way it is.

Consider the forearm holding a weight. The elbow joint is the pivot. The bicep muscle attaches very close to the elbow — typically about 5 cm from the joint. The hand holding a weight is about 35 cm from the elbow. This is a third-class lever: the muscle effort is between the pivot and the load. The mechanical advantage is $5/35$, about 0.14 — meaning the muscle must exert about seven times the weight being held. This seems inefficient, but the arrangement provides a large range of motion and high speed of movement at the hand, which matters far more in daily life than raw force amplification.

The spine, hip, knee, and ankle joints all involve similar lever arrangements. Understanding these geometries helps physical therapists design rehabilitation programs, ergonomists design safe workplaces, and sports scientists optimize athletic technique.

Bridge Note *Biomechanics courses at university extend these torque analyses to three-dimensional motion, using vectors and matrix methods. The concepts here — lever arm, moment equation, mechanical advantage — are the exact foundation for that work.*

20.11 Torque in Structures and Machines

Engineers apply the principles of static equilibrium constantly. Every bridge, crane, shelf bracket, and roof truss must satisfy both equilibrium conditions under all expected loads — otherwise it deforms or collapses.

In a simple beam bridge, the weight of the bridge itself and the vehicles on it are distributed loads that create torques about the support points. Engineers must calculate the reaction forces at each support, which depend on where the load is located. A heavy truck near one end of a bridge produces much larger forces at that end's support than at the other — a consequence of the torque equations.

Cranes use counterweights specifically to manage torque. The arm of a crane holding a load creates a large clockwise torque about the crane's central column. A counterweight on the opposite side of the column creates a counterclockwise torque to balance it. Without the counterweight, the crane would tip over — an expensive and dangerous example of rotational disequilibrium.

Bolted flanges, gear trains, screws, and turbine blades all involve torque calculations. The concepts introduced here scale directly to industrial applications.

20.12 Common Mistakes in Torque Problems

Students working through torque and equilibrium problems repeatedly make the same errors. Being aware of these pitfalls can save considerable time and frustration.

- Using the distance to the point of application rather than the perpendicular distance from the pivot to the line of action of the force. Always use the lever arm — the perpendicular distance.
- Forgetting the weight of the beam or plank itself. A uniform beam has its weight acting at its geometric center. This torque must be included in $\Sigma\tau = 0$.
- Using the wrong pivot. Remember: you can choose any pivot, but choosing it at the location of an unknown force simplifies the algebra enormously.
- Inconsistent sign conventions. Define clockwise or counterclockwise as positive at the start and use that definition throughout.
- Applying only one equilibrium condition. Both $\Sigma F = 0$ and $\Sigma\tau = 0$ are required. Using only one usually leaves unknowns unsolvable.
- Treating torque and energy as the same because they share units ($\text{N}\cdot\text{m}$). They are different quantities. Torque is not energy.

Chapter Summary

Torque is the rotational effect of a force applied at a distance from a pivot. It depends on force magnitude, lever arm length, and the angle of application: $\tau = F \cdot r \cdot \sin \theta$. The unit of torque is the newton-meter (N·m).

The lever arm is the perpendicular distance from the pivot to the line of action of the force — not merely the distance to the point where the force is applied.

The principle of moments: the net torque about any pivot equals the algebraic sum of all individual torques.

For static equilibrium, two conditions must both be satisfied: (1) net force = 0 in every direction (translational equilibrium), and (2) net torque = 0 about any chosen pivot (rotational equilibrium).

A well-chosen pivot eliminates unknown forces from the torque equation by giving them a zero lever arm.

Levers come in three classes based on the relative positions of pivot, effort, and load. Mechanical advantage = lever arm of effort / lever arm of load.

The human body is a system of third-class levers optimized for range of motion and speed rather than force amplification. Engineering structures rely on equilibrium analysis to remain safe under load.

Key Terms

Term	Definition
Torque (τ)	The rotational effect of a force applied at a distance from a pivot; $\tau = F \cdot r \cdot \sin \theta$, measured in N·m.
Pivot (fulcrum)	The fixed point about which an object rotates or tends to rotate.
Lever arm (moment arm)	The perpendicular distance from the pivot to the line of action of the force.
Line of action	An imaginary line extending along the direction of a force in both directions.
Principle of moments	The net torque about any pivot equals the algebraic sum of all individual torques.
Static equilibrium	The state in which an object is at rest and has zero net force and zero net torque.

Translational equilibrium	The condition $\Sigma F = 0$; the net force in every direction is zero.
Rotational equilibrium	The condition $\Sigma \tau = 0$; the net torque about any chosen pivot is zero.
Mechanical advantage	The ratio of output force to input force in a simple machine; for levers, $MA = d_{\text{effort}}/d_{\text{load}}$.
First-class lever	A lever in which the pivot lies between the effort and the load (e.g., seesaw).
Second-class lever	A lever in which the load lies between the pivot and the effort (e.g., wheelbarrow).
Third-class lever	A lever in which the effort lies between the pivot and the load (e.g., forearm, tweezers).

Concept Review Questions

Answer the following in complete sentences. No formulas are required unless the question asks you to use them.

13. In your own words, why does torque depend on the lever arm rather than simply the distance to where a force is applied?
14. Explain why a force directed exactly toward the pivot produces no torque. Use the concept of lever arm in your answer.
15. What are the two conditions for static equilibrium? Explain why both are necessary by giving an example where one condition is met but the other is not.
16. Why is the choice of pivot point important when solving torque problems? What makes a particular pivot a good choice?
17. A wrench is used to tighten a bolt. Explain how you would apply the force to maximize the torque delivered to the bolt.
18. Classify the following as first-, second-, or third-class levers: (a) a seesaw, (b) a wheelbarrow, (c) tweezers, (d) a nutcracker, (e) the forearm curling a weight.
19. Explain why the muscles of the human arm must exert forces several times greater than the objects the hand holds. What is the trade-off that makes this design advantageous?
20. A crane uses a counterweight on the opposite side of its arm from the load. Explain using torque principles why the counterweight is necessary.

Torque Problems

21. A 30 N force is applied perpendicular to a wrench 0.20 m from the bolt. Calculate the torque.
22. The same 30 N force is applied at 45° to the wrench. Calculate the torque and compare it to the previous result.
23. A uniform 4.0 kg beam, 2.0 m long, is pivoted at its left end. A 20 N downward force is applied at the right end. Calculate the net torque about the pivot.
24. A seesaw is 5.0 m long with the pivot at its center. A 40 kg child sits 1.5 m from the pivot on the left. Where must a 30 kg child sit on the right side to balance the seesaw?
25. A uniform 10 kg plank is 3.0 m long and supported at both ends. A 60 N downward force is applied 1.0 m from the left end. Find the upward reaction forces at each support.
26. A ladder of mass 15 kg and length 4.0 m leans against a smooth wall at 60° to the ground. Find the torque exerted by gravity about the base of the ladder.
27. A bolt requires 25 N·m of torque to tighten. A mechanic applies a perpendicular force with a 0.30 m wrench. What force is required? How much would the required force decrease if a 0.45 m wrench were used instead?

Static Equilibrium Problems

28. A uniform 8.0 kg beam, 3.0 m long, is supported at its left end by a hinge and at its right end by a vertical cable. A 50 N weight hangs 1.0 m from the right end. Find (a) the tension in the cable and (b) the reaction force at the hinge.
29. A 70 kg person stands on a uniform plank of mass 20 kg. The plank is 4.0 m long and rests on two supports 0.5 m from each end. The person stands 1.2 m from the left support. Find the reaction forces at each support.
30. A 12 kg sign hangs from a horizontal rod of negligible mass. The rod is 0.80 m long, attached to a wall at one end by a hinge, and supported by a cable at the other end at 35° above horizontal. Find the cable tension and the components of the hinge force.
31. A uniform door (mass 18 kg, width 0.90 m) is supported by two hinges located 0.20 m from the top and 0.20 m from the bottom. Find the forces exerted by each hinge if the door is in equilibrium.

Lever and Machine Problems

32. A crowbar is 1.2 m long. The fulcrum is 0.10 m from the load end. What mechanical advantage does this crowbar provide? If you apply a 150 N force, how large a load can you lift?
33. A wheelbarrow has a load of 250 N applied 0.40 m from the wheel axle. The handles are 1.60 m from the axle. What force must be applied at the handles to lift the load?

34. The bicep muscle attaches 4.0 cm from the elbow joint. A person holds a 15 N weight with the forearm horizontal; the hand is 32 cm from the elbow. What force must the bicep exert? What class of lever is this?

Chapter 20 Checkpoint

Before moving on to Chapter 21, you should be able to do the following without looking at your notes.

I can...	See Section
Define torque and state the three factors it depends on	20.2
Explain the lever arm and distinguish it from the distance to the application point	20.3
Calculate torque using $\tau = F \cdot r \cdot \sin \theta$	20.4
Apply the principle of moments to find unknown forces or distances	20.5
State both conditions for static equilibrium	20.6
Choose an effective pivot point in an equilibrium problem	20.7
Solve static equilibrium problems using free-body diagrams	20.8
Calculate mechanical advantage for each class of lever	20.9
Identify lever classes in the human body and explain their design trade-offs	20.10
Explain how torque analysis applies to engineering structures	20.11
Identify and avoid the most common mistakes in torque problems	20.12

Continue to Chapter 21: Center of Mass and Stability →